

Public Finance

BIEM – Spring 2026

**Tax inefficiencies and their implications
for Optimal Taxation (Ch. 20)**



Introduction

1. Taxation and Economic Efficiency

2. Optimal Commodity Taxation

3. Optimal Income Taxes



Taxation and Economic Efficiency 1

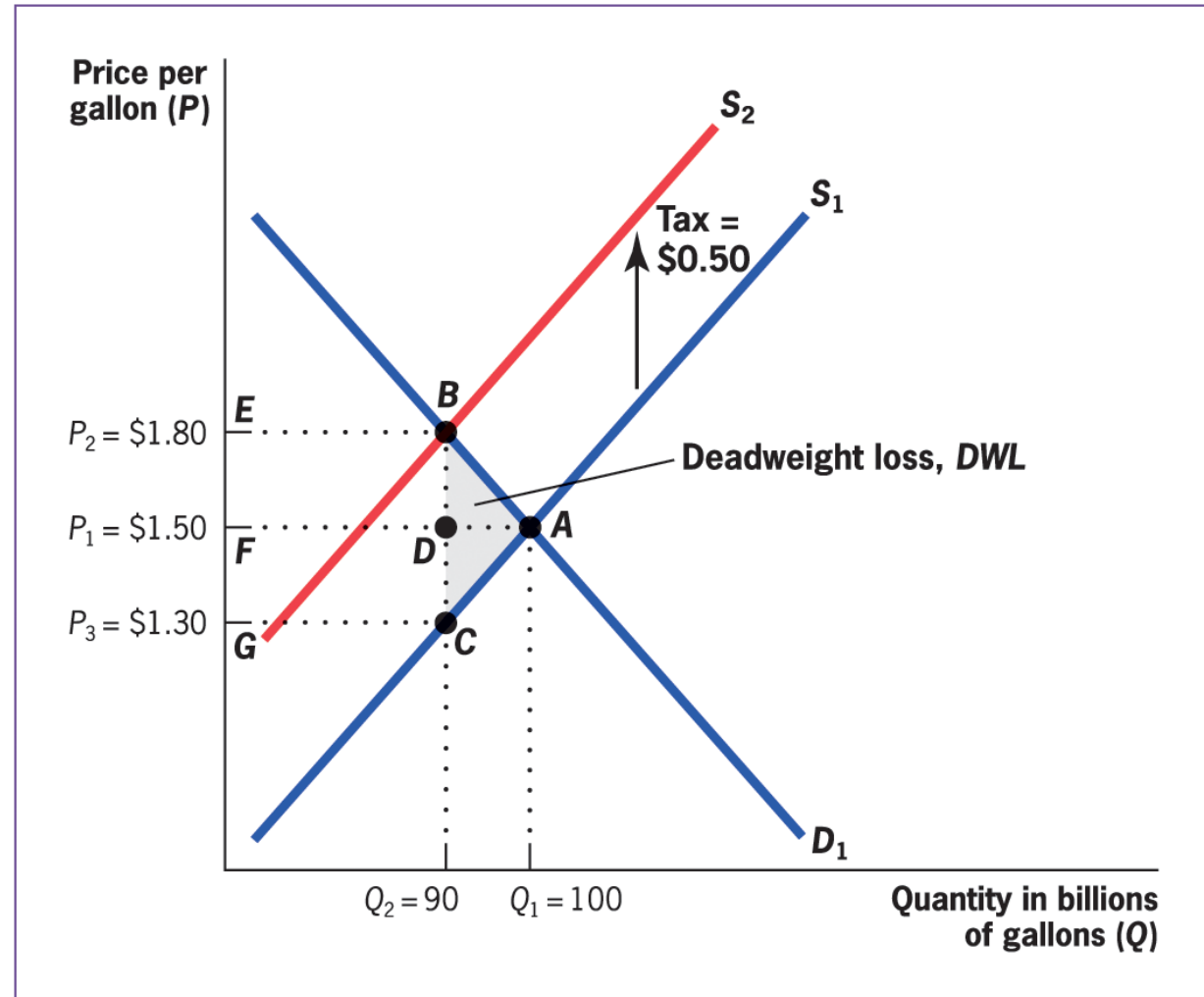
- Usually, the market produces efficient outcomes.
- Taxes interfere in the market and reduce efficiency.
- People substitute away from the taxed product, using less efficient alternatives.
- Some taxes have much larger efficiency costs than others.



Taxation and Economic Efficiency 2

Initial equilibrium: point **A**

Impact of 50¢ per gallon tax levied on producers of gasoline, causing them to **decrease the quantity supplied** at each price



Taxation and Economic Efficiency 3

New equilibrium: point **B**

The **DWL** arises because there are trades ($Q_1 - Q_2$) for which social marginal benefits (demand curve) exceed social marginal costs (supply curve) that are not made

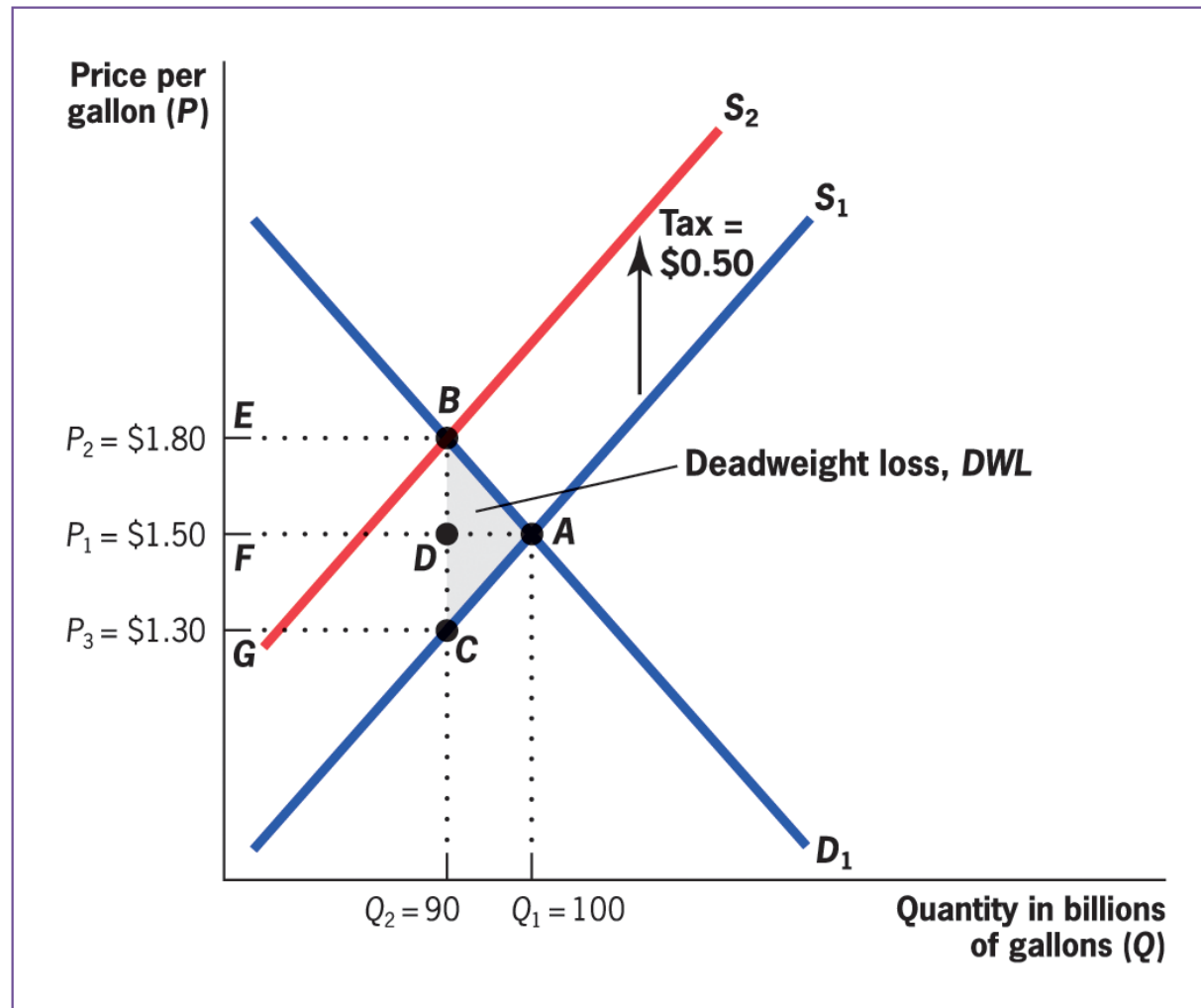


Figure 20-1

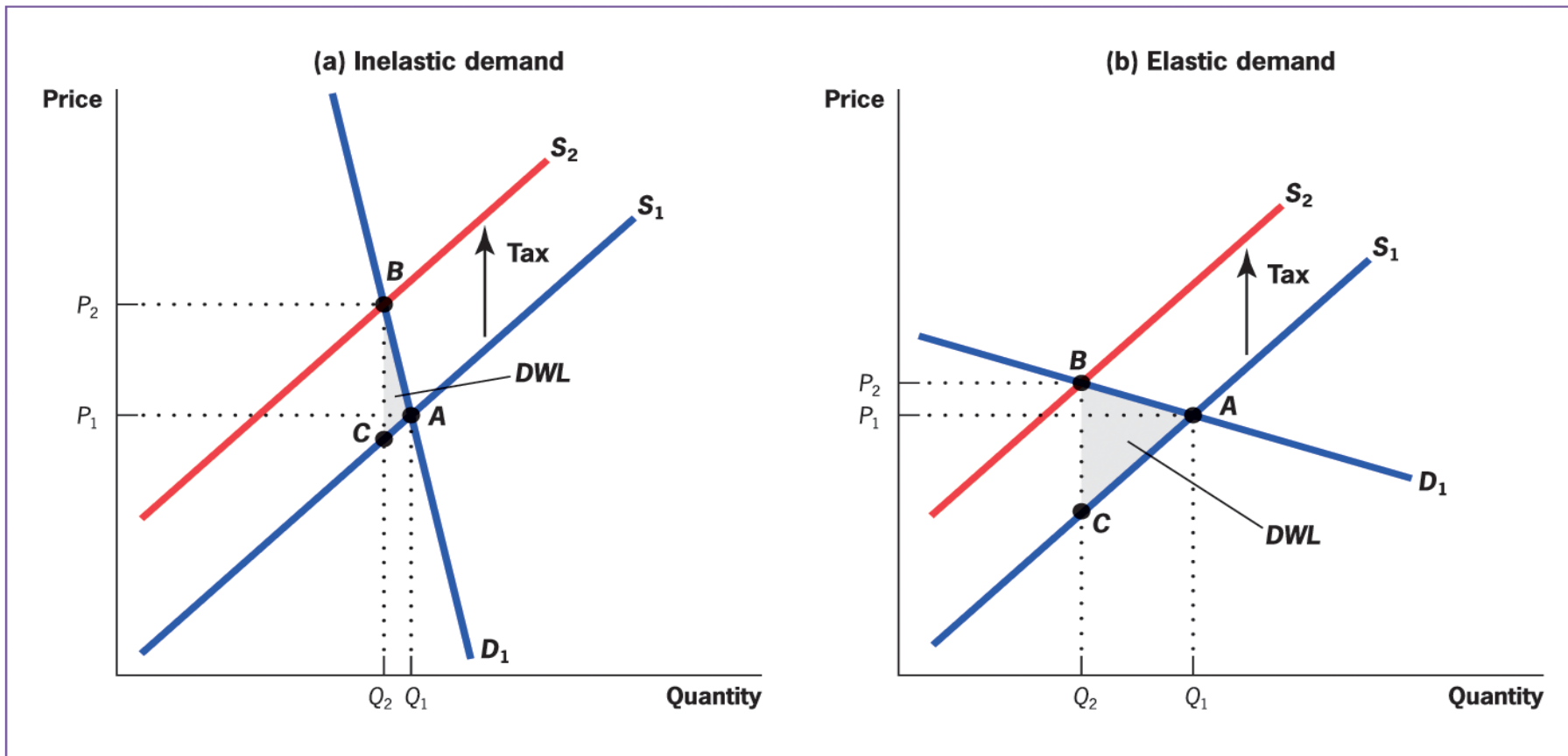
Taxation and Economic Efficiency 4

- The **DWL** measures the inefficiency of taxation: the amount of consumer and producer surplus society loses by imposing a tax
- DWL is determined by changed in quantities when a tax is imposed because this change captures the number of socially efficient trades that are not being made
- The outcome of this efficiency analysis would be the same **if the tax were imposed on consumers** instead of producers
- The more elastic is demand or supply, the larger the DWL.



Elasticities determine Tax Inefficiency

Figure 20-2



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As demand and supply elasticities rise, the **DWL** of taxation grows

Empirical Evidence: The Window Tax 1

In 1696, English King William III needed money to finance an ongoing war with France.

- To raise this revenue, the king imposed a **new tax** based on the **number of windows in a home**, which was an indicator of a home's value
- The problem was that individuals could minimize their tax by boarding up or otherwise removing windows. This rendered the tax a less efficient measure of home value and created a **deadweight loss** by making it less pleasant (and therefore of less value to residents) to live in their homes



Empirical Evidence: The Window Tax 2

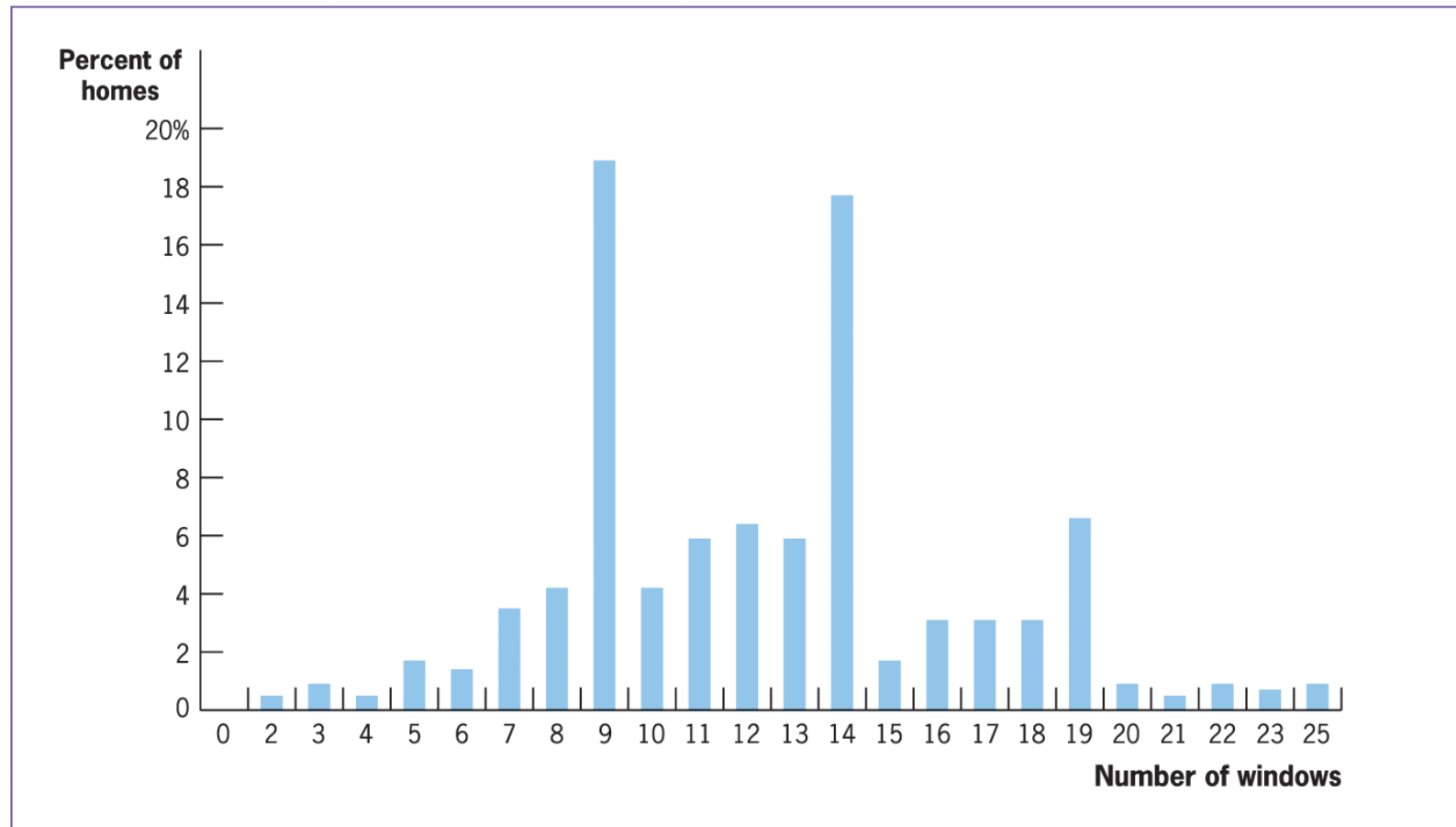
Oates and Schwab (2015) presented a test for whether this was the case: they looked at **whether the number of windows responded** to “notches” in the tax schedule.

- From 1747 to 1757:
 - there was no tax if the house had fewer than ten windows,
 - a tax of 6 pence per window for 10 to 14 windows;
 - a tax of 9 pence per window if there were 15 to 19 windows;
 - a tax of 1 shilling (12 pence) per window for 20 windows or more.
- The marginal tax rate jumped enormously on building the 10th, 15th, or 20th window.



Empirical Evidence: The Window Tax 3

Figure 20-3



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There are obvious spikes at 9, 14, and 19 windows, exactly what we would predict from the effects of this tax.



Empirical Evidence: The Window Tax 4

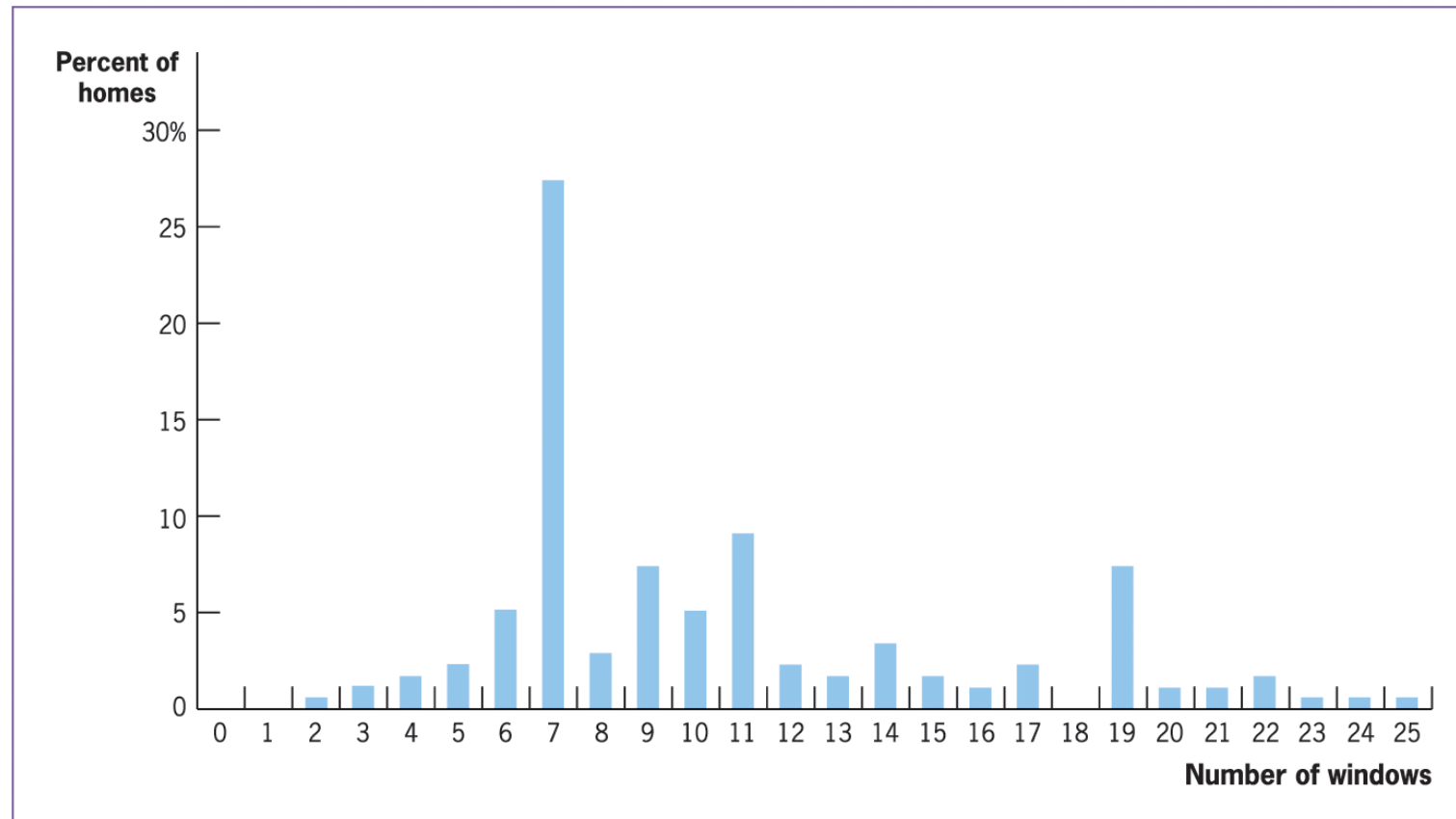
One could argue that perhaps there was some British custom or some other reason for having 9, 14, or 19 windows in a house.

- Oates and Schwab address this point by using the fact that in 1761, the government added a provision that houses with 8 or 9 windows would pay 1 shilling per window.
- This means that after 1761, there was suddenly an incentive to keep the number of windows to 7 or fewer. This is exactly what happened.



Empirical Evidence: The Window Tax

Figure 20-4



In data from 1761 to 1765, the authors find that 27% of houses have 7 windows, while only 5% have 6 windows, and 3% have 8 windows. Clearly, this was a response to the tax.



APPLICATION: Tax Avoidance in Practice

Keynes: “The avoidance of taxes is the only pursuit that still carries any reward.”
Some examples of avoidance:

1. The Papal States taxed salt heavily, so Tuscan bakers stopped using it. Even today, Tuscan bread is saltless.
2. In the early 1980s, Cyprus’s building tax applied to finished structures. Homeowners put steel bars jutting out from their roofs to avoid the tax.
3. The U.S. government taxes imported shoes at a much higher rate than imported slippers. Some shoe manufacturers pass their products off as slippers, e.g., by implanting a temporary layer of felt in the soles.
4. Alberobello’s trulli



APPENDIX: Deriving the formula for Deadweight Loss

$$DWL = -\frac{1}{2} \times \underbrace{\Delta Q}_{\text{Change in Q induced by the tax}} \times \underbrace{\tau}_{\text{Size of the tax}}$$

$$\eta_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$



$$\frac{\Delta Q}{Q} = \eta_s \times \frac{\Delta P}{P}$$

$$\eta_d = \frac{\Delta Q}{\Delta P + \tau} \times \frac{P}{Q}$$



$$\Delta P = \frac{\eta_d}{(\eta_s - \eta_d)} \times \tau$$

$$\Delta Q = \frac{\eta_s \eta_d}{(\eta_s - \eta_d)} \times \tau \frac{Q}{P}$$

Tax induced change in quantity



The Mathematics of Optimal Taxation: Deriving the formula for Deadweight Loss

$$DWL = -\frac{1}{2} \times \Delta Q \times \tau$$

$$\Delta Q = \frac{\eta_s \eta_d}{(\eta_s - \eta_d)} \times \tau \frac{Q}{P}$$

$$DWL = -\frac{\eta_s \eta_d}{2(\eta_s - \eta_d)} \times \tau^2 \frac{Q}{P}$$



Determinants of Deadweight Loss

$$DWL = -\frac{\eta_s \eta_d}{2(\eta_s - \eta_d)} \times \tau^2 \frac{Q}{P}$$

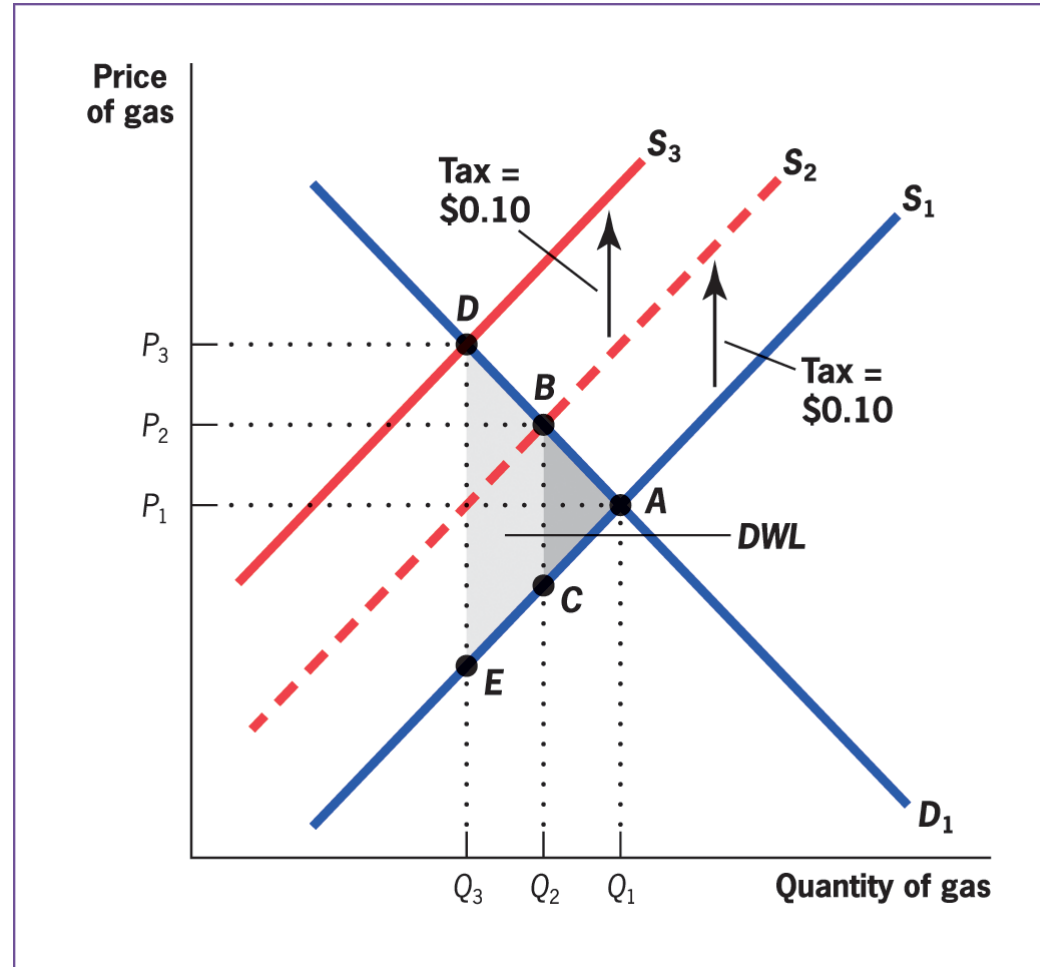
- η_s and η_d are the elasticity of supply and demand, τ is the tax rate, and Q and P are the quantity and price.
- DWL rises with the elasticities of demand and supply
- DWL rises with the square of the tax, so marginal DWL rises with the tax rate.
 - **Marginal deadweight loss:** The increase in deadweight loss per unit increase in the tax.



Determinants of Deadweight Loss

Figure 20-5

- Initial equilibrium: **A**
- First tax of 10¢ per unit on producers
- Second tax of 10¢ per unit on producers



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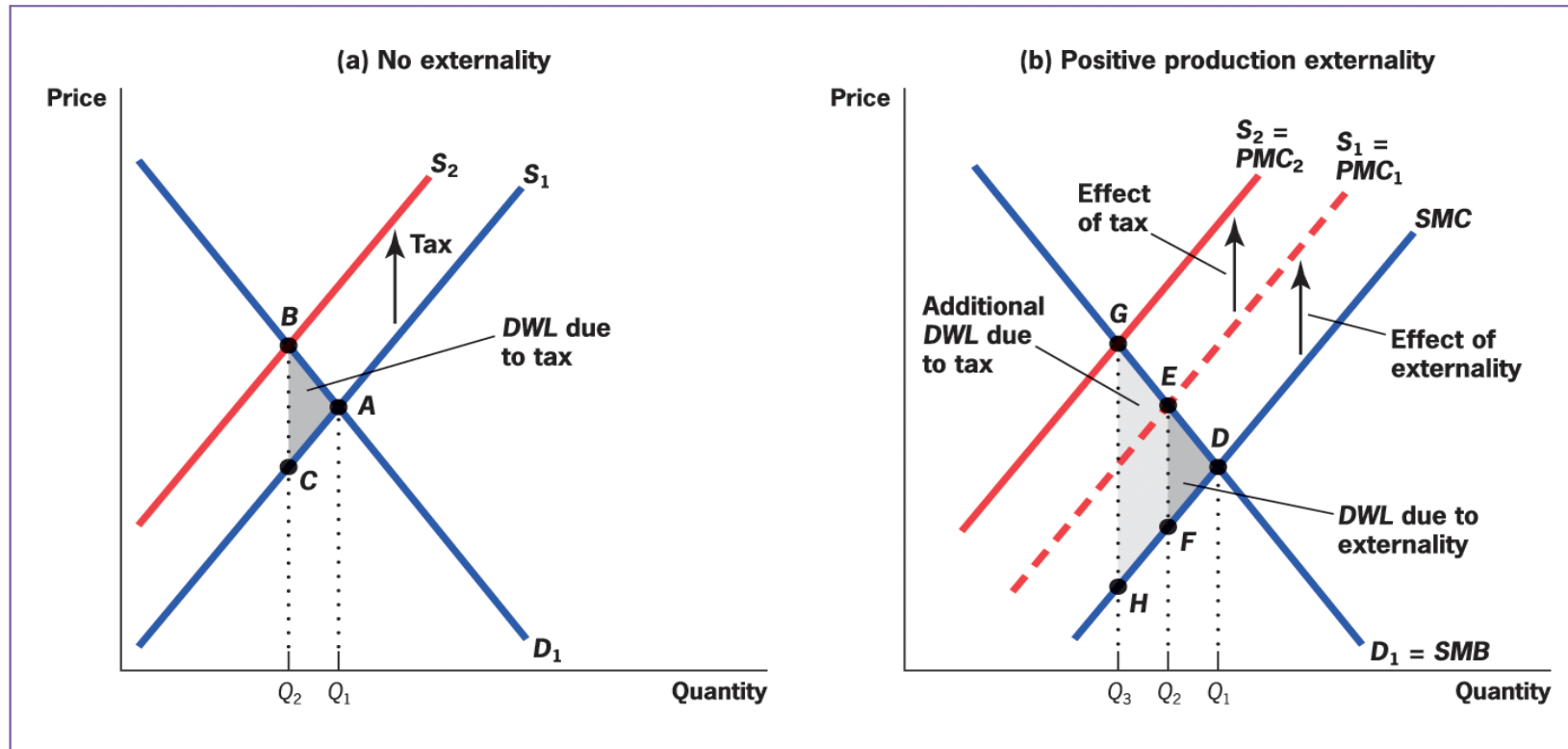


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The marginal deadweight loss rises disproportionately with the tax rate.

A Tax System's Efficiency Is Affected by a Market's Preexisting Distortions

Figure 20-6



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a) A tax in a market with **no preexisting distortions** creates a deadweight loss equal to triangle BAC.

b) A **positive externality** in the market creates a deadweight loss triangle EDF; imposing a tax on this market results in a deadweight loss that is larger by the area of the trapezoid GEFH.

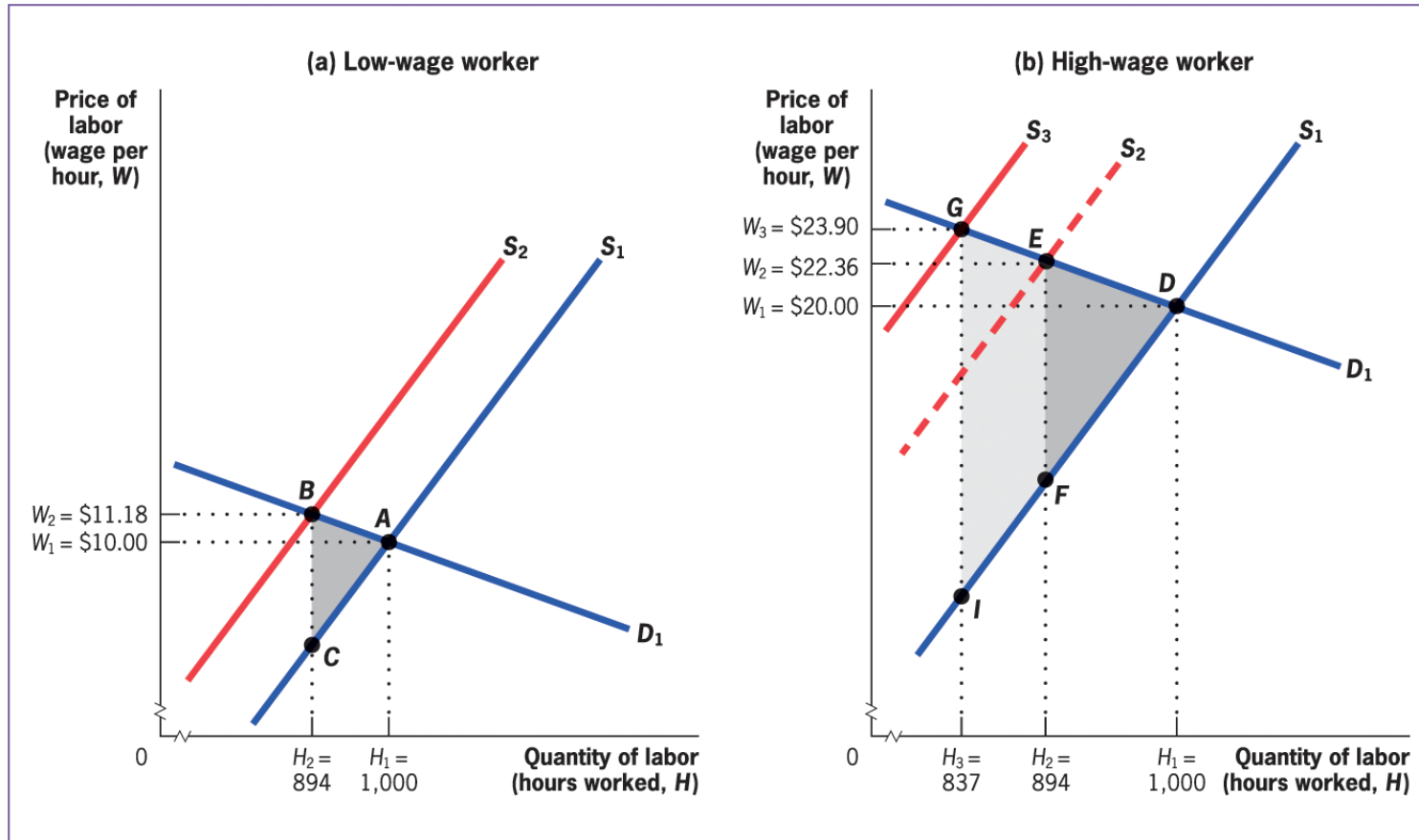
Progressive Tax Systems Can Be Less Efficient

- Previous insights about DWL apply not only to taxation of goods, but also to **taxation of income**
- There can be large efficiency costs to moving from a proportional to a **progressive tax system**
- **Example:**
 - Suppose there are two people, one with a wage of \$10 per hour and one with a wage of \$20 per hour.
 - For both, a 10% rise in wages leads them to supply 10% more labor (elasticity of labor supply = 1).
 - Elasticity of labor demand is also 1.



Progressive Tax Systems Can Be Less Efficient: **proportional**

Figure 20-7

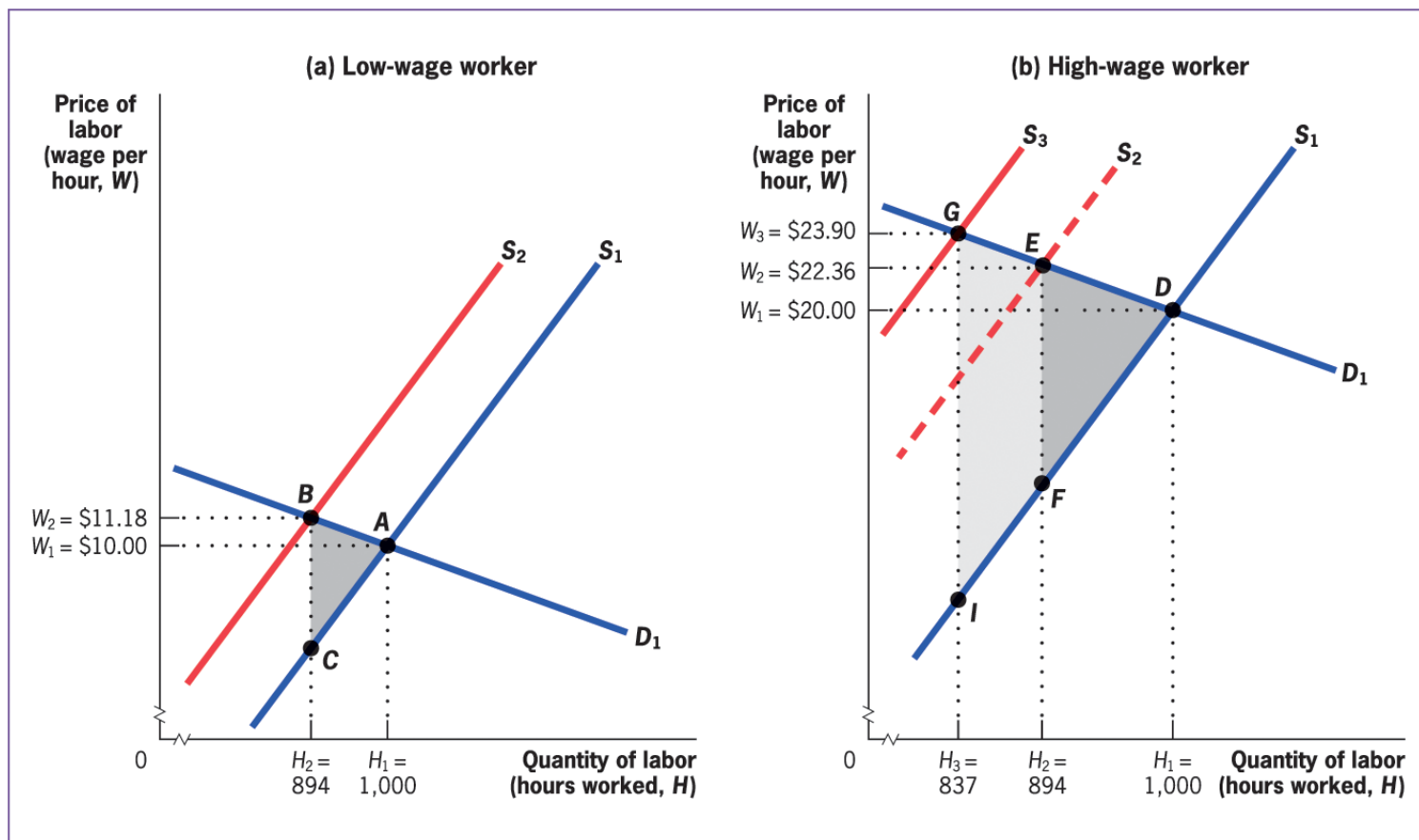


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Initially, the government imposes an equal tax on the low-wage worker and the high-wage worker, resulting in deadweight losses shown by triangles BAC and EDF.

Progressive Tax Systems Can Be Less Efficient: **progressive**

Figure 20-7



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When system is replaced with no tax on low-wage workers, there is no DWL for this worker, but the DWL for the high-wage worker increases by the trapezoid GEFI. This results in an increase in deadweight loss because additional taxes must be collected from high-wage workers in order to collect the same revenues as before.



Governments Should “Smooth” Tax Rates over Time

- Government efficiency in taxation over time is maximized by tax smoothing, by having a relatively constant tax rate over time rather than high taxes in some periods and low taxes in others.
- High and then low tax rates produce a larger DWL than steady medium tax rates.



Optimal Commodity Taxation

- In the previous section we looked at the *positive* question of how to measure DWL
- This section turns to the *normative* question of how the existence of a DWL should be taken into account in the design of the tax system
 - Commodity taxation (taxation of goods)
 - Taxation of income



Ramsey Taxation: The Theory of Commodity Taxation

- **Optimal commodity taxation:** Choosing the tax rates across goods to minimize deadweight loss for a given government revenue requirement.
- **Ramsey Rule:** To minimize the DWL of a tax system while raising a fixed amount of revenue, taxes should be set across commodities so that the ratio of the marginal deadweight loss to marginal revenue raised is equal across commodities:

$$\frac{MDWL_i}{MR_i} = \lambda$$

- **Value of additional government revenues:** The value of having another dollar in the government's hands relative to its next best use in the private sector.



Inverse Elasticity Rule

- If supply is infinitely elastic (as in perfect competition), the Ramsey Rule becomes:

$$\tau_i^* = -\frac{1}{\eta_i} \times \lambda$$

- τ_i^* is the optimal tax, η_i is the elasticity of demand, and λ is some constant.
- Optimal taxation therefore balances two rules:
 - Elasticity rule: Lower taxes on goods with more elastic demand.
 - Broad base rule: Better to tax a wide variety of goods moderately than a few goods heavily.



Equity Implications of the Ramsey Model

Imagine that the government had only two goods it could tax: cereal and caviar:

- The elasticity of demand for caviar is much higher than that for cereal.
- The inverse elasticity rule suggests that the government should tax cereal much more highly than caviar.
- This means taxing more heavily the good consumed by poor people, worsening vertical equality.



Optimal Income Taxes

- Most tax revenue in the United States and other developed countries is from income taxes
- **Optimal income taxation:** Choosing the tax rates across income groups to **maximize social welfare** subject to a government revenue requirement
- Social welfare function guides the trade-off between progressivity and efficiency



Conclusion

- The fundamental issue in designing tax policy is the equity-efficiency trade-off.
- Tax efficiency comes down to two key principles:
 - The more elastically supplied or demanded the good, the larger the deadweight loss from the tax.
 - The higher the tax rate, the larger the incremental deadweight loss of taxation.



Exercises



MC 1

What determines the size of the deadweight loss from a tax?

- a) The statutory incidence of the tax
- b) The level of tax revenue raised
- c) The elasticities of supply and demand
- d) The number of taxpayers subject to the tax



MC 2

Why does the marginal deadweight loss of taxation rise more than proportionally with the tax rate?

- a) Because tax revenue increases linearly with the tax rate
- b) Because the tax base shrinks faster at higher tax rates
- c) Because elasticities increase with income
- d) Because the deadweight loss triangle grows with the square of the tax rate



MC 3

Why is the economic incidence of a tax independent of whether it is levied on consumers or producers?

- a) Because statutory incidence does not affect prices
- b) Because demand is always perfectly inelastic
- c) Because prices adjust to equate supply and demand
- d) Because tax revenue is fixed



MC 4

According to the inverse elasticity rule, which goods should be taxed more heavily?

- a) Goods with elastic demand
- b) Goods with inelastic demand
- c) Luxury goods
- d) Goods consumed by high-income individuals



MC 5

Why can progressive tax systems be less efficient than proportional systems?

- a) Because they always raise less revenue
- b) Because they eliminate redistribution
- c) Because they concentrate taxation on high-elasticity margins
- d) Because they reduce average tax rates



True/False (please provide a justification)

- 1) Imposing a tax on a market with a positive externality creates a larger deadweight loss than taxing a market without distortions.
- 2) Tax smoothing over time reduces deadweight loss relative to highly variable tax rates.
- 3) The Ramsey rule always leads to equitable tax outcomes.



Exercises

EXERCISE 1

Market demand: $Q = 240 - 6P$

Market supply: $Q = -40 + 2P$

- Find equilibrium price and quantity without taxation.
- Calculate the deadweight loss of a tax of \$4 per unit.
- Explain whether DWL changes if the tax is levied on consumers instead of producers.



EXERCISE 2

The government of Washlovia imposes a per-unit tax of the same size on clothes dryers in two regions, East and West Washlovia.

- In East Washlovia, the demand elasticity is -2.4 .
- In West Washlovia, the demand elasticity is -1.7 .

Assume that supply is perfectly elastic in both regions.

- a) In which region does the tax lead to a larger reduction in quantity sold? Explain.
- b) In which region is the deadweight loss larger? Explain using the elasticity concept.
- c) In which region does the tax raise more revenue? Briefly justify your answer.